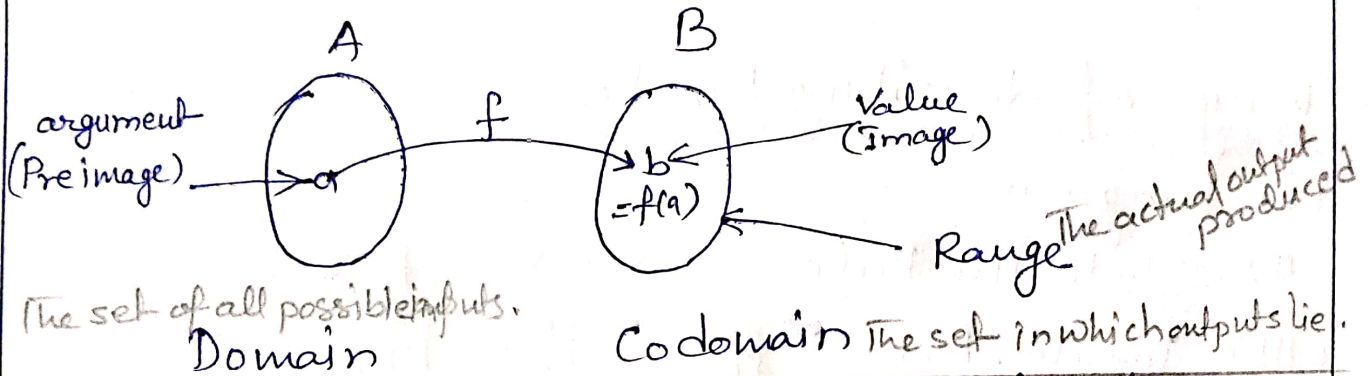


Functions

Def: - Let A and B be non-empty sets. A relation from A to B such that to every element $a \in A$ there corresponds a unique element $b \in B$ is called a function f from A to B .

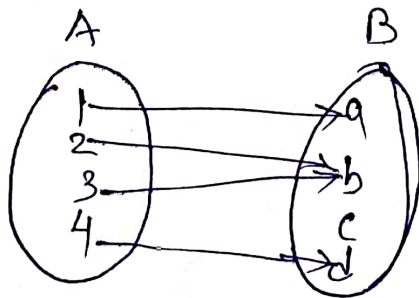


Note (1) In a function there cannot be any element in A from which two (or more) arrows go and Also there cannot be any element in B from which no arrow goes.
 A में कोई एक ही बाहर और A का element एक से ज्यादा के साथ Related हो

Ex-1 Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$ and

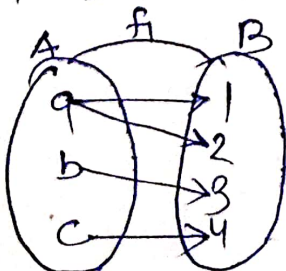
Let $f = \{(1, a), (2, b), (3, b), (4, d)\}$

Here we have $f(1) = a, f(2) = b, f(3) = b, f(4) = d$

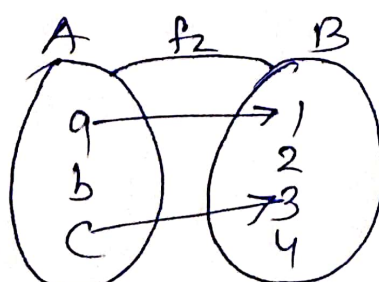


Ex-2 Let $A = \{a, b, c\}$, $B = \{1, 2, 3, 4\}$ and let

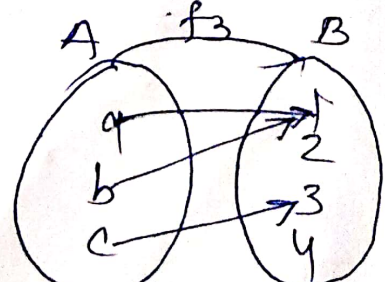
$f_1 = \{(a, 1), (a, 2), (b, 3), (c, 4)\}$ and $f_2 = \{(a, 1), (c, 3)\}$, $f_3 = \{(a, 1), (b, 1), (c, 3)\}$



f_1 is not function



f_2 is not function



f_3 is a function.

Ex:- (i) $f(x) = x^2$ is a function

each input x gives a single output x^2

(ii) the relation $y^2 = x$ is not a function if seen as $x \rightarrow y$. because one input $x = 4$ gives two outputs ($y = 2$ and $y = -2$).

Ways of describing functions

(i) Diagram

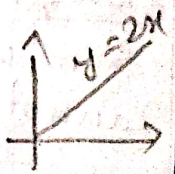
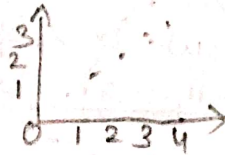
(ii) State the Rule $f: X \rightarrow Y$ & $f(x) = 2x$

(iii) Tabular form

x	-2	1	3	4
$f(x)$	8	2	7	9

(iv) Ordered Pair $f = \{(-2, 8), (1, 2), (3, 7), (4, 9)\}$

(v) Graph of a function.



Types of functions

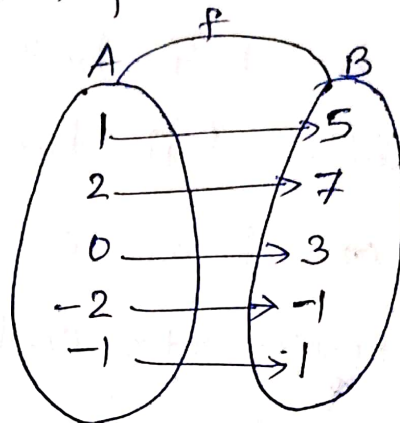
(1) One-to-one (Injective) function

Each element of the domain maps to a unique element in the codomain.

A function $f: X \rightarrow Y$ is one-to-one if $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$ or $f(x_1) = f(x_2)$ implies $x_1 = x_2$.

Ex:- Let $f(x) = (2x+3)$

x	1	2	0	-2	-1
$f(x)$	5	7	3	-1	1



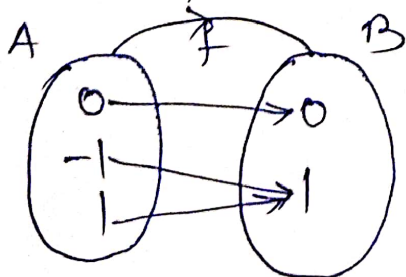
(2) On to (Surjective) function

Every element of the codomain has at least one pre-image in the domain.

OR
A function $f: A \rightarrow B$ is called surjective or onto if every element of $b \in B$ is an image of at least one element a of A .

Ex:- Let $A = \{0, -1, 1\}$ and $B = \{0, 1\}$ Let $A \rightarrow B$ where $f(a) = |a|$. Is f onto?

We have $f(0) = |0| = 0$, $f(-1) = |-1| = 1$, $f(1) = |1| = 1$



Hence f is onto function.

(3) One to one and On to (Bijective) function

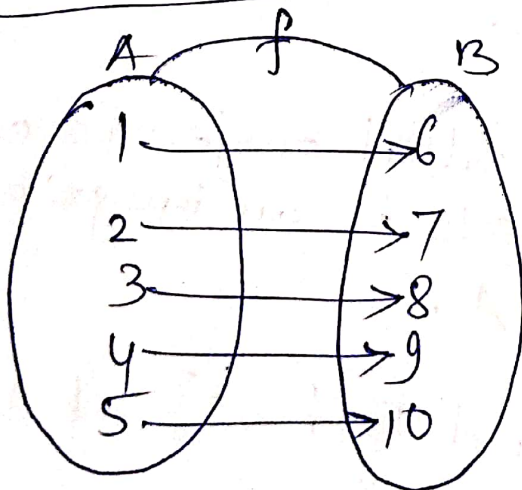
The function which is both injective and surjective.

OR
Each input has one unique output and every output is matched.

OR
If $f: A \rightarrow B$ is both one-to-one and onto then f is called bijective. Such function is also called one to one correspondence between A & B .

Ex:- $f(x) = x + 5$ (when domain and codomain are all real numbers).

x	1	2	3	4	5
$f(x)=y$	6	7	8	9	10



Many One function

Two or more different elements of the domain map to the same element in the codomain.

Ex: - $f(x) = x^2$. Here $f(2) = 4$ and $f(-2) = 4$.

Constant function

Every input maps to the same output

Ex: - $f(x) = 5$ for all x , output is always 5.

Identity function

When output is equal to input.

Ex: - $f(x) = x$.

Inverse function

The inverse function of a function f is another function usually denoted by f^{-1} that 'undoes' the action of f .

- If $f: A \rightarrow B$ then $f^{-1}: B \rightarrow A$.
- $f[f^{-1}(x)] = x$ and $f^{-1}[f(x)] = x$.

Ex: let $f(x) = 2x + 3$ then f^{-1} will be.

$$\text{let } y = 2x + 3 \quad \therefore y - 3 = 2x \quad \therefore \frac{y - 3}{2} = x$$

$$\text{So } f^{-1}(x) = \frac{x - 3}{2}$$

check $f[f^{-1}(x)] = 2\left[\frac{x - 3}{2}\right] + 3 = x$

$$f^{-1}[f(x)] = \frac{(2x + 3) - 3}{2} = x$$

Q-1 Are the following functions (i) Injective (ii) Surjective?

(a) $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = x^2 - 4x$$

For Injective if $f(x_1) = f(x_2)$ then $x_1 = x_2$

Let two elements x_1 and x_2 in $\mathbb{R}$ and let $f(x_1) = f(x_2)$

$$\therefore x_1^2 - 4x_1 = x_2^2 - 4x_2$$

$$(x_1^2 - x_2^2) - 4(x_1 - x_2) = 0$$

$$(x_1 - x_2)(x_1 + x_2) - 4(x_1 - x_2) = 0$$

$$(x_1 - x_2)[x_1 + x_2 - 4] = 0$$

$$\therefore x_1 - x_2 = 0 \text{ or } x_1 + x_2 - 4 = 0$$

$$\therefore x_1 = x_2 \text{ or } x_1 = 4 - x_2 \therefore x_1 \neq x_2 \text{ always}$$

$\therefore f$ is not injective.

For bijective - every element $y (= f(x))$ in Y we must have at least one corresponding element x in X .

Now $y = x^2 - 4x$, since we cannot solve this equation for x , we try Positive, zero and negative values of y . Geometrically this means, we try whether all lines parallel to the x -axis intersect the curve.

$$\text{If } y = 6 \text{ (say) then } x^2 - 4x - 6 = 0 \therefore x = \frac{4 \pm \sqrt{16 + 24}}{2} \therefore x \in \mathbb{R}$$

$$\text{If } y = 0 \text{ (say) then } x^2 - 4x = 0 \therefore x(x - 4) = 0 \therefore x = 0 \text{ or } x = 4 \therefore x \in \mathbb{R}$$

$$\text{If } y = -6 \text{ (say) then } x^2 - 4x + 6 = 0, \therefore x = \frac{4 \pm \sqrt{16 - 24}}{2} \text{ imaginary } \therefore x \notin \mathbb{R}$$

$\therefore$ for every $y \in Y$ there is no $x \in X$

$\therefore f$ is not surjective.

(b) $g: \mathbb{R} - \{1\} \rightarrow \mathbb{R} - \{2\}$
 $g(x) = \frac{2x-3}{x-1}$

for Injective let $g(x_1) = g(x_2)$

$$\therefore \frac{2x_1-3}{x_1-1} = \frac{2x_2-3}{x_2-1}$$

$$(2x_1-3)(x_2-1) = (2x_2-3)(x_1-1)$$

$$\therefore 2x_1x_2 - 2x_1 - 3x_2 + 3 = 2x_1x_2 - 2x_2 - 3x_1 + 3$$

$$\therefore x_1 - x_2 = 0 \quad \therefore x_1 = x_2$$

$\therefore g$ is injective

Now for surjective let $y = \frac{2x-3}{x-1}$

$$\therefore y(x-1) = 2x-3$$

$$\therefore xy - y = 2x - 3$$

$$\therefore xy - 2x = y - 3$$

$$\therefore x(y-2) = y-3$$

$$x = \frac{y-3}{y-2}$$

If $y \neq 2$ for every y there corresponds $x \in \mathbb{R}$

$\therefore g$ is surjective in $\mathbb{R} - \{2\}$

$\therefore g$ is bijective and $g^{-1}(x) = \frac{x-3}{x-2}$